Topic 08: Laplace Transform of Derivatives
Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

Contents

1 Opening Hook

Hook

You have been building a machine. Topic by topic, you assembled the definition, the existence conditions, the standard table, the properties. Now comes the moment the machine proves its worth. Take any ODE with initial conditions — say $y'' - 3y' + 2y = e^{4t}$, $y(0) = 1$, $y'(0) = 0$. Apply the Laplace Transform to both sides. Every derivative turns into a simple multiplication by s. The ODE becomes a linear algebraic equation. Solve for $Y(s)$. Then invert. Done. No characteristic equations, no undetermined coefficients. Just algebra. This is why engineers love the Laplace Transform.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this...

Ordinary differential equations with initial conditions are the bread-and-butter of engineering analysis. Every vibrating structure, every electrical circuit, every control system is governed by ODEs. Classical methods — characteristic equation, variation of parameters, method of undetermined coefficients — are cumbersome when the forcing functions are discontinuous, impulsive, or piecewise. The Laplace Transform *automates* the handling of initial conditions; they fall out of the algebra automatically. No more hunting for particular solutions and patching them with homogeneous solutions.

Mistake	**Consequence**
Forgetting the initial condition terms in $\mathcal{L}\{f'\}$	Losing the IC contribution and getting wrong ODE solutions
Applying the formula without checking differentiability	Getting incorrect results for non-smooth functions
Not tracking which initial condition goes where in $\mathcal{L}\{f''\}$	Swapping $f(0)$ and $f'(0)$ roles — wrong final answer

Key Takeaway

The Laplace Transform converts an initial-value problem (IVP) into a purely algebraic problem. Initial conditions are *automatically built into* the transformed equation — there is no separate step to impose them.

3 You Already Know This (Intuition First)

Recall Fourier analysis: in the frequency domain of electrical engineering, differentiation with respect to time becomes multiplication by $j\omega$. This is exactly the same idea the Laplace Transform exploits. The Laplace Transform takes the derivative $f'(t)$ and replaces it with $sF(s)$ (minus an initial condition correction). It is not magic — it follows directly from integration by parts applied to the definition integral

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt.$$

Think of s as a "frequency-like" variable. Multiplying by s in the s-domain is the algebraic counterpart of differentiating once in the t-domain. The initial condition term $f(0)$ is the "memory" of the system at the starting instant — it cannot be ignored.

4 Transform of the First Derivative

Theorem: $\mathcal{L}\{f'(t)\}$

Statement. Let $f(t)$ be continuous on $[0, \infty)$ and of exponential order, and let $f'(t)$ be piecewise continuous on $[0, \infty)$. Then

$$\boxed{\mathcal{L}\{f'(t)\} = s\,F(s) - f(0)}$$

where $F(s) = \mathcal{L}\{f(t)\}$.

Proof by integration by parts.
Start from the definition:

$$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st}\, f'(t)\, dt.$$

Choose $u = e^{-st}$ and $dv = f'(t)\, dt$, so $du = -s\, e^{-st}\, dt$ and $v = f(t)$. Integration by parts gives:

$$\int_0^\infty e^{-st}\, f'(t)\, dt = \Big[e^{-st}\, f(t)\Big]_0^\infty - \int_0^\infty (-s)\, e^{-st}\, f(t)\, dt.$$

Evaluate the boundary term. Because $f(t)$ is of exponential order there exists $M > 0$ and σ_0 such that $|f(t)| \le Me^{\sigma_0 t}$ for all large t. For $\mathrm{Re}(s) > \sigma_0$:

$$\lim_{t\to\infty} e^{-st}\, f(t) \le \lim_{t\to\infty} M\, e^{-(s-\sigma_0)t} = 0.$$

At $t = 0$ the boundary term evaluates to $-e^0\, f(0) = -f(0)$. Therefore:

$$\mathcal{L}\{f'(t)\} = \big(0 - f(0)\big) + s\int_0^\infty e^{-st}\, f(t)\, dt = -f(0) + s\,F(s) = s\,F(s) - f(0).$$

What Did We Learn?

$\mathcal{L}\{f'(t)\} = s\,F(s) - f(0)$. Differentiation in the t-domain $\longleftrightarrow$ multiplication by s in the s-domain, minus the value of the function at $t = 0$.

5 Transform of the Second Derivative

Theorem: $\mathcal{L}\{f''(t)\}$

Statement.

$$\boxed{\mathcal{L}\{f''(t)\} = s^2\,F(s) - s\,f(0) - f'(0)}$$

Derivation. Treat $f''(t) = (f')'(t)$ and apply the first derivative formula twice.
Let $g(t) = f'(t)$, so $g'(t) = f''(t)$.
Apply $\mathcal{L}\{g'(t)\} = s\,G(s) - g(0)$:

$$\mathcal{L}\{f''(t)\} = s\,\mathcal{L}\{f'(t)\} - f'(0).$$

Now substitute $\mathcal{L}\{f'(t)\} = s\,F(s) - f(0)$:

$$\mathcal{L}\{f''(t)\} = s\big[s\,F(s) - f(0)\big] - f'(0) = s^2\,F(s) - s\,f(0) - f'(0).$$

What Did We Learn?

Each additional derivative contributes one more power of s and peels off one more initial condition. $\mathcal{L}\{f''\} = s^2F(s) - s\,f(0) - f'(0)$.

6 General nth Derivative Formula

General Formula

Theorem. If $f, f', \dots, f^{(n-1)}$ are continuous on $[0,\infty)$ and of exponential order, and $f^{(n)}$ is piecewise continuous, then

$$\boxed{\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \cdots - s\, f^{(n-2)}(0) - f^{(n-1)}(0)}$$

Proof by induction.
Base case ($n = 1$): already proved — $\mathcal{L}\{f'\} = sF(s) - f(0)$. ✓
Inductive step: assume the formula holds for $n = k$, i.e.,

$$\mathcal{L}\{f^{(k)}(t)\} = s^k F(s) - s^{k-1} f(0) - \cdots - f^{(k-1)}(0).$$

For $n = k+1$, write $f^{(k+1)}(t) = \big(f^{(k)}\big)'(t)$ and apply the first-derivative formula with $h(t) = f^{(k)}(t)$:

$$\mathcal{L}\{f^{(k+1)}(t)\} = s\,\mathcal{L}\{f^{(k)}(t)\} - f^{(k)}(0).$$

Substitute the inductive hypothesis:

$$= s\big[s^k F(s) - s^{k-1} f(0) - \cdots - f^{(k-1)}(0)\big] - f^{(k)}(0) = s^{k+1} F(s) - s^k f(0) - \cdots - f^{(k)}(0).$$

Explicit $n = 3$ case:

$$\begin{aligned}\mathcal{L}\{f'''(t)\} &= s\,\mathcal{L}\{f''(t)\} - f''(0)\\ &= s\big[s^2 F(s) - s\, f(0) - f'(0)\big] - f''(0)\\ &= s^3 F(s) - s^2 f(0) - s\, f'(0) - f''(0).\end{aligned}$$

n	$\mathcal{L}\{f^{(n)}(t)\}$
1	$s\,F(s) - f(0)$
2	$s^2 F(s) - s\, f(0) - f'(0)$
3	$s^3 F(s) - s^2 f(0) - s\, f'(0) - f''(0)$
4	$s^4 F(s) - s^3 f(0) - s^2 f'(0) - s\, f''(0) - f'''(0)$

What Did We Learn?

The pattern is crystal clear: power of s decreases by 1 for each successive initial condition $f^{(k)}(0)$. For an nth-order ODE, applying $\mathcal{L}$ produces an algebraic equation of degree n in s — solvable by elementary algebra.

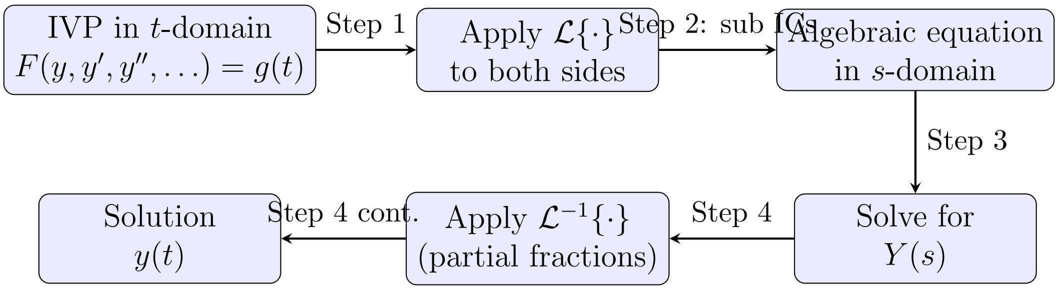
7 The ODE Solution Workflow

4-Step Workflow for Solving IVPs via Laplace Transforms

Step 1: Apply $\mathcal{L}\{\cdot\}$ to *both sides* of the ODE. Use the derivative formulas to replace y', y'', etc. by algebraic expressions in $Y(s) = \mathcal{L}\{y(t)\}$.
Step 2: Substitute the numerical values of the initial conditions $y(0)$, $y'(0)$, ...
Step 3: Collect and isolate $Y(s)$ — solve the resulting linear algebraic equation

for $Y(s)$.
Step 4: Apply the inverse Laplace transform $y(t) = \mathcal{L}^{-1}\{Y(s)\}$ using partial fractions and the standard Laplace table to recover $y(t)$.

4-Step Workflow — Block Diagram:



What Did We Learn?

The four steps are mechanical: transform → substitute ICs → solve algebra → invert. Each step is elementary on its own; the power comes from chaining them.

8 Worked Examples

Example 1 — First-Order ODE: $y' + 2y$

Step 1: Apply $\mathcal{L}$ to both sides.
$\mathcal{L}\{y' + 2y\} = \mathcal{L}\{4\}$
Using the first-derivative formula and linearity:

$$\big[s\,Y(s) - y(0)\big] + 2\,Y(s) = \frac{4}{s}.$$

Step 2: Substitute the initial condition $y(0) = 1$.

$$s\,Y(s) - 1 + 2\,Y(s) = \frac{4}{s}.$$

Step 3: Collect $Y(s)$.

$$(s+2)\,Y(s) = \frac{4}{s} + 1 = \frac{4+s}{s}.$$

$$Y(s) = \frac{s+4}{s(s+2)}.$$

Step 4: Partial fractions.
Write $\dfrac{s+4}{s(s+2)} = \dfrac{A}{s} + \dfrac{B}{s+2}$.
Multiply both sides by $s(s+2)$:

$$s + 4 = A(s+2) + B\,s.$$

Set $s = 0$: $4 = 2A \Rightarrow A = 2$.

Set $s = -2$: $2 = -2B \Rightarrow B = -1$.
Therefore:

$$Y(s) = \frac{2}{s} - \frac{1}{s+2}.$$

Step 5: Invert.

$$y(t) = \mathcal{L}^{-1}\left\{\frac{2}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = 2 - e^{-2t}.$$

Verification: $y(0) = 2 - 1 = 1$ ✓. $y'(t) = 2e^{-2t}$. $y' + 2y = 2e^{-2t} + 2(2 - e^{-2t}) = 2e^{-2t} + 4 - 2e^{-2t} = 4$ ✓.

What Did We Learn? (Example 1)

For a first-order IVP, the Laplace method reduces to: one formula application, one partial-fraction split, one table look-up. The initial condition $y(0) = 1$ was automatically incorporated in Step 2.

Example 2 — Second-Order ODE: $y'' - 3y' + 2y$

Step 1: Apply $\mathcal{L}$ to both sides.

$$\mathcal{L}\{y''\} - 3\,\mathcal{L}\{y'\} + 2\,\mathcal{L}\{y\} = 0.$$

Expand each Laplace term:

$$\big[s^2Y - s\,y(0) - y'(0)\big] - 3\big[s\,Y - y(0)\big] + 2\,Y = 0.$$

Step 2: Substitute $y(0) = 1$, $y'(0) = 0$.

$$(s^2Y - s - 0) - 3(s\,Y - 1) + 2\,Y = 0.$$

$$s^2Y - s - 3s\,Y + 3 + 2Y = 0.$$

Step 3: Collect $Y(s)$.

$$(s^2 - 3s + 2)\,Y = s - 3.$$

Factor: $s^2 - 3s + 2 = (s-1)(s-2)$.

$$Y(s) = \frac{s-3}{(s-1)(s-2)}.$$

Step 4: Partial fractions.
Write $\dfrac{s-3}{(s-1)(s-2)} = \dfrac{A}{s-1} + \dfrac{B}{s-2}$.
Multiply both sides by $(s-1)(s-2)$:

$$s - 3 = A(s-2) + B(s-1).$$

Set $s = 1$: $-2 = -A \Rightarrow A = 2$.
Set $s = 2$: $-1 = B \Rightarrow B = -1$.

Therefore:

$$Y(s) = \frac{2}{s-1} - \frac{1}{s-2}.$$

Step 5: Invert.

$$y(t) = 2e^t - e^{2t}.$$

Verification: $y(0) = 2 - 1 = 1$ ✓. $y'(t) = 2e^t - 2e^{2t}$; $y'(0) = 2 - 2 = 0$ ✓. $y'' - 3y' + 2y = (2e^t - 4e^{2t}) - 3(2e^t - 2e^{2t}) + 2(2e^t - e^{2t}) = (2-6+4)e^t + (-4+6-2)e^{2t} = 0$ ✓.

What Did We Learn? (Example 2)

A homogeneous second-order IVP is solved without guessing exponential solutions. The characteristic roots $s = 1$ and $s = 2$ appear naturally as the poles of $Y(s)$.

Example 3 — Second-Order ODE with Forcing: $y'' + y$

Step 1: Apply $\mathcal{L}$ to both sides.
Note $\mathcal{L}\{\cos t\} = \dfrac{s}{s^2+1}$.

$$\left[s^2 Y - s\,y(0) - y'(0)\right] + Y = \frac{s}{s^2+1}.$$

Step 2: Substitute $y(0) = 1$, $y'(0) = 0$.

$$s^2 Y - s + Y = \frac{s}{s^2+1}.$$

$$(s^2+1)\,Y = s + \frac{s}{s^2+1} = \frac{s(s^2+1)+s}{s^2+1} = \frac{s^3+2s}{s^2+1}.$$

Step 3: Solve for $Y(s)$.

$$Y(s) = \frac{s^3+2s}{(s^2+1)^2}.$$

Step 4: Partial fractions over repeated irreducible quadratic.
Write:

$$\frac{s^3+2s}{(s^2+1)^2} = \frac{As+B}{s^2+1} + \frac{Cs+D}{(s^2+1)^2}.$$

Multiply both sides by $(s^2+1)^2$:

$$s^3 + 2s = (As+B)(s^2+1) + (Cs+D).$$

Expand the right-hand side:

$$= As^3 + As + Bs^2 + B + Cs + D = As^3 + Bs^2 + (A+C)s + (B+D).$$

Match coefficients:

$$\begin{aligned} s^3 &: A = 1, \\ s^2 &: B = 0, \\ s^1 &: A + C = 2 \Rightarrow C = 1, \\ s^0 &: B + D = 0 \Rightarrow D = 0. \end{aligned}$$

Therefore:

$$Y(s) = \frac{s}{s^2+1} + \frac{s}{(s^2+1)^2}.$$

Step 5: Invert.
$\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} = \cos t.$
For $\mathcal{L}^{-1}\left\{\frac{s}{(s^2+1)^2}\right\}$, use the known result $\mathcal{L}\left\{\frac{t\sin t}{2}\right\} = \frac{s}{(s^2+1)^2}$ (derivable via the $-dF/ds$ rule).
Therefore:

$$y(t) = \cos t + \tfrac{1}{2}\, t \sin t.$$

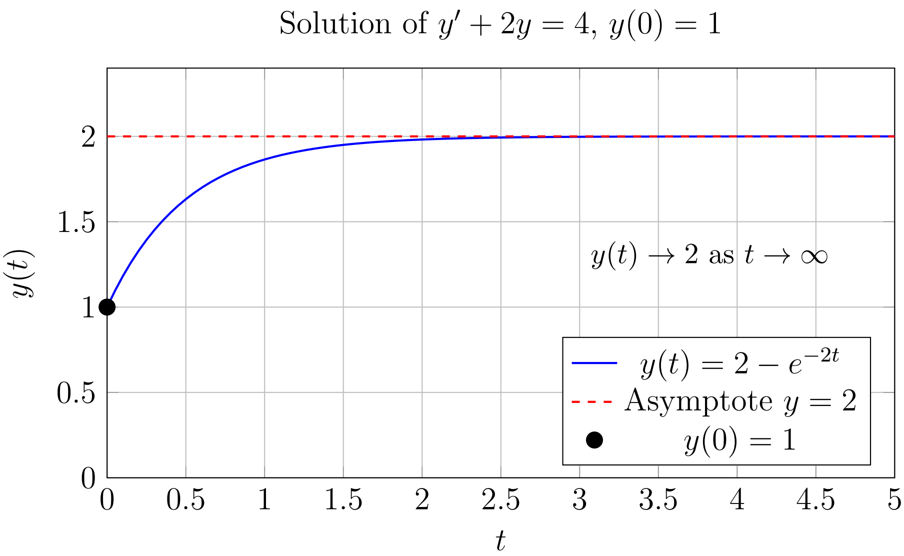
Verification: $y(0) = 1 + 0 = 1$ ✓. $y'(0) = (-\sin t + \frac{1}{2}\sin t + \frac{1}{2}t\cos t)\big|_{t=0} = 0 + 0 + 0 = 0$ ✓.

What Did We Learn? (Example 3)

Even when the forcing function $\cos t$ shares the natural frequency of the system ($\omega = 1$), the Laplace method handles the resonance algebraically: it appears as a repeated pole at $s = \pm i$ and produces the $t \sin t$ term that is the hallmark of resonance.

9 pgfplots Visualization

Figure: The solution $y(t) = 2 - e^{-2t}$ to Example 1, showing the initial value $y(0) = 1$ and the asymptote $y = 2$.



Caption: The Laplace method automatically incorporates the initial condition $y(0) = 1$ — no need to impose it separately. The solution rises smoothly from 1 toward the steady state 2.

10 Excel Example (Numerical Verification)

We numerically verify $y(t) = 2 - e^{-2t}$ against an Euler-method integration of $y' = 4 - 2y$, $y(0) = 1$, using step size $h = 0.1$.

Col A t	**Col B (Exact)** $y_{\text{exact}} = 2 - e^{-2t}$	**Col C (Euler)** y_{Euler}	**Col D (Error)** $\lvert B - C \rvert$	**Excel Formu**
0.0	1.000000	1.000000	0.000000	A2=0; B2==2-EXP(-2*A2); C2=
0.1	1.181269	1.200000	0.018731	A3==A2+0.1; B3==2-EXP(-2*A3); C3==C2+
0.2	1.329680	1.360000	0.030320	A4==A3+0.1; B4==2-EXP(-2*A4); C4==C3+
0.3	1.451188	1.488000	0.036812	(Drag formulas down f

Setup instructions:

1. Enter 0 in A2. In A3 type `=A2+0.1` and fill down to row 32 (covering $t = 0$ to $t = 3$).
2. In B2 type `=2-EXP(-2*A2)` and fill down — this is the Laplace-derived exact formula.
3. In C2 type 1 (initial condition). In C3 type `=C2+0.1*(4-2*C2)` (Euler step: $y_{n+1} = y_n + h(4 - 2y_n)$) and fill down.
4. In D2 type `=ABS(B2-C2)` and fill down — this is the absolute error at each step.

Sample values at selected times:

t	Exact y	Euler y	Error
0.0	1.000000	1.000000	0.000000
0.5	1.632121	1.590938	0.041183
1.0	1.864665	1.836660	0.028005
2.0	1.981684	1.971015	0.010669
3.0	1.997521	1.993735	0.003786

What Did We Learn? (Excel)

The Euler approximation converges to the Laplace-derived exact solution $2 - e^{-2t}$ as t grows. The error is largest near $t = 0.3$ and diminishes as both curves approach the steady state $y = 2$. This confirms that the algebraic Laplace solution is correct.

11 Python Example

Listing 1: Solving IVPs with SymPy Laplace transform and dsolve, plus numerical verification

```
import sympy as sp
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

```

```python
# ------------------------------------------------------------------
# Part 1: Symbolic solution using sympy.dsolve
# ------------------------------------------------------------------
t = sp.Symbol('t', positive=True)
s = sp.Symbol('s')
y = sp.Function('y')

# ODE: y'' - 3*y' + 2*y = 0, y(0)=1, y'(0)=0
ode = y(t).diff(t, 2) - 3*y(t).diff(t) + 2*y(t)
sol = sp.dsolve(ode, y(t), ics={y(0): 1, y(t).diff(t).subs(t, 0):
    0})
print("dsolve solution:", sol)
# Expected output: Eq(y(t), 2*exp(t) - exp(2*t))

# ------------------------------------------------------------------
# Part 2: Manual Laplace approach using sympy
# ------------------------------------------------------------------
F = sp.Function('F')
Y = sp.Symbol('Y')
y0, dy0 = 1, 0  # initial conditions

# L{y''} = s^2 Y - s*y(0) - y'(0)
# L{y'}  = s Y - y(0)
# L{y}   = Y
# ODE in s-domain: (s^2 - 3s + 2)*Y - (s*y0 + dy0) + 3*y0 = 0
# => (s^2-3s+2)*Y = s*y0 - 3*y0 + dy0 = s - 3
Ys_expr = (s*y0 - 3*y0 + dy0) / (s**2 - 3*s + 2)
print("Y(s) =", sp.factor(Ys_expr))
# Expected output: Y(s) = (s - 3)/((s - 1)*(s - 2))

# Partial fractions
pf = sp.apart(Ys_expr, s)
print("Partial fractions:", pf)
# Expected output: -1/(s - 2) + 2/(s - 1)

# Inverse Laplace
yt = sp.inverse_laplace_transform(Ys_expr, s, t)
print("y(t) =", yt)
# Expected output: y(t) = 2*exp(t) - exp(2*t)

# ------------------------------------------------------------------
```

```
# Part 3: Numerical verification with scipy odeint
#
    ------------------------------------------------------------------
def system(Y_vec, t_val):
    y_val, dy_val = Y_vec
    d2y = 3*dy_val - 2*y_val    # from y'' = 3y' - 2y
    return [dy_val, d2y]

t_num = np.linspace(0, 2, 200)
Y0 = [1, 0]   # y(0)=1, y'(0)=0
sol_num = odeint(system, Y0, t_num)

# Exact solution
y_exact = 2*np.exp(t_num) - np.exp(2*t_num)

# Max absolute error
max_err = np.max(np.abs(sol_num[:, 0] - y_exact))
print(f"Max numerical error vs Laplace solution: {max_err:.6e}")
# Expected output: Max numerical error vs Laplace solution: ~1e
    -06

# Plot
plt.figure(figsize=(8,4))
plt.plot(t_num, y_exact, 'b-', label='Laplace: $2e^t - e^{2t}$')
plt.plot(t_num, sol_num[:, 0], 'r--', label='odeint numerical')
plt.xlabel('t')
plt.ylabel('y(t)')
plt.title("ODE: $y'' - 3y' + 2y = 0$, y(0)=1, y'(0)=0")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig('ode_comparison.png', dpi=150)
print("Plot saved as ode_comparison.png")
```

What Did We Learn? (Python)

`sympy.inverse_laplace_transform` confirms the algebraic result $y(t) = 2e^t - e^{2t}$. The `odeint` numerical solution matches to within $\sim 10^{-6}$, validating both the Laplace derivation and the numerical integrator. The workflow — define ODE symbolically, apply Laplace, do partial fractions, invert — mirrors the pencil-and-paper method exactly.

12 Viva-Style Oral Questions

1. **Q: State and prove the formula for $\mathcal{L}\{f'(t)\}$.**

   A: $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$. The proof uses integration by parts on $\int_0^\infty e^{-st} f'(t)\,dt$ with $u = e^{-st}$, $dv = f'(t)\,dt$. The boundary term at ∞ vanishes because $f(t)$ is of exponential order, giving $-f(0) + sF(s)$.

2. **Q: Why does $f(0)$ appear in the formula for $\mathcal{L}\{f'(t)\}$ and not $f(\infty)$?**

   A: The boundary term from integration by parts is $\left[e^{-st}f(t)\right]_0^\infty$. The upper limit gives $\lim_{t\to\infty} e^{-st}f(t) = 0$ (exponential decay dominates). The lower limit gives $-e^0 f(0) = -f(0)$. So only the value at $t = 0$ survives.

3. **Q: Derive $\mathcal{L}\{f''(t)\}$ from $\mathcal{L}\{f'(t)\}$.**

   A: Write $f''(t) = (f')'(t)$. Let $g = f'$. Then $\mathcal{L}\{g'(t)\} = sG(s) - g(0) = s\mathcal{L}\{f'(t)\} - f'(0)$. Substituting $\mathcal{L}\{f'\} = sF(s) - f(0)$: $\mathcal{L}\{f''\} = s[sF(s) - f(0)] - f'(0) = s^2F(s) - sf(0) - f'(0)$.

4. **Q: Write the general formula for $\mathcal{L}\{f^{(n)}(t)\}$.**

   A: $\mathcal{L}\{f^{(n)}\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$. Each successive initial condition is multiplied by one lower power of s.

5. **Q: Describe the 4-step workflow for solving an IVP using Laplace transforms.**

   A: (1) Apply $\mathcal{L}$ to both sides of the ODE — each derivative becomes an algebraic expression in $Y(s)$. (2) Substitute numerical initial conditions. (3) Collect $Y(s)$ and solve the resulting linear algebraic equation. (4) Apply $\mathcal{L}^{-1}$ via partial fractions and the standard table to get $y(t)$.

6. **Q: Why does the Laplace method handle initial conditions automatically?**

   A: Because the derivative formulas $\mathcal{L}\{f'\} = sF(s) - f(0)$, $\mathcal{L}\{f''\} = s^2F(s) - sf(0) - f'(0)$, etc., contain $f(0)$, $f'(0)$, $\ldots$ explicitly. When you substitute their numerical values in Step 2, those values are already embedded in the algebraic equation for $Y(s)$. There is no separate "apply IC" phase at the end.

7. **Q: What does $Y(s) = \mathcal{L}\{y(t)\}$ represent?**

   A: $Y(s)$ is the Laplace transform of the unknown solution $y(t)$. It is a rational function of s (for polynomial-coefficient ODEs). Its poles are the roots of the characteristic polynomial of the ODE, and its partial-fraction expansion encodes the entire time-domain solution.

8. **Q: Under what conditions is the formula $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$ valid?**

   A: (i) $f(t)$ must be continuous on $[0, \infty)$. (ii) $f(t)$ must be of exponential order (so that $e^{-st}f(t) \to 0$ as $t \to \infty$). (iii) $f'(t)$ must be at least piecewise continuous on $[0, \infty)$. If f has a jump discontinuity, the formula must be modified to account for the jump.

13 Descriptive Questions

1. **Derive $\mathcal{L}\{f'(t)\}$ from the definition of the Laplace transform using integration by parts. State clearly all conditions required on f and f'.**

   Model answer: Starting from $\int_0^\infty e^{-st}f'(t)\,dt$, apply integration by parts with $u = e^{-st}$, $dv = f'(t)\,dt$. The boundary term is $[e^{-st}f(t)]_0^\infty$; the upper limit vanishes under exponential-order and $\mathrm{Re}(s) > \sigma_0$, leaving $-f(0)$. The remaining integral is $s\int_0^\infty e^{-st}f(t)\,dt = sF(s)$. Hence $\mathcal{L}\{f'\} = sF(s) - f(0)$. Conditions: f continuous, f' piecewise continuous, both of exponential order.

2. **Starting from $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$, derive $\mathcal{L}\{f''(t)\}$ step by step. Show every substitution explicitly.**

   Model answer: Let $g = f'$, so $g' = f''$. Apply $\mathcal{L}\{g'\} = sG(s) - g(0) = s\mathcal{L}\{f'\} - f'(0)$. Substitute: $= s(sF(s) - f(0)) - f'(0) = s^2F(s) - sf(0) - f'(0)$.

3. **Solve the first-order IVP $y' + 3y = 6$, $y(0) = 2$ using the Laplace transform method. Show all four steps.**

   Model answer: Step 1: $(sY - 2) + 3Y = 6/s$. Step 2: already substituted. Step 3: $(s + 3)Y = 6/s + 2 = (6 + 2s)/s$; $Y = (6 + 2s)/[s(s + 3)]$. Partial fractions: $A/s + B/(s + 3)$; $A = 2$, $B = 0$. So $Y = 2/s$. Step 4: $y(t) = 2$. Check: $y' = 0$; $0 + 6 = 6$ ✓; $y(0) = 2$ ✓.

4. **Solve the second-order IVP $y''+4y'+4y = 0$, $y(0) = 1$, $y'(0) = -2$ completely using Laplace transforms.**

   Model answer: Apply $\mathcal{L}$: $(s^2Y - s+2)+4(sY-1)+4Y = 0$. $(s^2+4s+4)Y = s+2$. $(s+2)^2Y = s+2 \Rightarrow Y = 1/(s+2)$. $y(t) = e^{-2t}$. Check: $y(0) = 1$ ✓; $y'(t) = -2e^{-2t}$; $y'(0) = -2$ ✓.

5. **State the general nth derivative formula. Prove it for $n = 3$ by applying the first-derivative result twice more from the $n = 2$ formula.**

   Model answer: General: $\mathcal{L}\{f^{(n)}\} = s^nF - s^{n-1}f(0) - \cdots - f^{(n-1)}(0)$. For $n = 3$: write $f''' = (f'')'$ and apply $\mathcal{L}\{h'\} = s\mathcal{L}\{h\} - h(0)$ with $h = f''$. $\mathcal{L}\{f'''\} = s\mathcal{L}\{f''\} - f''(0) = s(s^2F - sf(0) - f'(0)) - f''(0) = s^3F - s^2f(0) - sf'(0) - f''(0)$.

14 Practice Problems

1. Solve $y' - y = e^t$, $y(0) = 0$.

   Hint: $\mathcal{L}\{e^t\} = 1/(s - 1)$. After transform: $(s - 1)Y = 1/(s - 1)$, so $Y = 1/(s - 1)^2$. **Answer:** $y(t) = t\,e^t$.

2. Solve $y'' + 4y = \sin(2t)$, $y(0) = 0$, $y'(0) = 0$.

   Hint: $\mathcal{L}\{\sin 2t\} = 2/(s^2 + 4)$. You get a repeated factor $(s^2 + 4)^2$ in $Y(s)$; use the standard result $\mathcal{L}^{-1}\{1/(s^2 + a^2)^2\} = (1/2a^3)(\sin at - at\cos at)$. **Answer:** $y(t) = \frac{1}{8}(\sin 2t - 2t\cos 2t)$.

3. Solve $y'' - y' - 6y = 0$, $y(0) = 2$, $y'(0) = -1$.

   Hint: Factor $s^2 - s - 6 = (s - 3)(s + 2)$. Partial fractions give $A/(s - 3) + B/(s + 2)$. **Answer:** $y(t) = e^{3t} + e^{-2t}$.

4. Solve $2y' + y = 1$, $y(0) = 1$.

   Hint: $2(sY - 1) + Y = 1/s$; $(2s + 1)Y = 2 + 1/s = (2s + 1)/s$; $Y = 1/s$. **Answer:** $y(t) = 1$.

5. Solve $y'' + 2y' + y = t$, $y(0) = 0$, $y'(0) = 0$.

   Hint: $Y = 1/[s^2(s + 1)^2]$. Use partial fractions with terms $A/s + B/s^2 + C/(s + 1) + D/(s + 1)^2$. **Answer:** $y(t) = t - 2 + (2 + t)e^{-t}$.

6. Solve $y''' - y = 0$, $y(0) = 1$, $y'(0) = 0$, $y''(0) = 1$.

   Hint: $\mathcal{L}\{y'''\} = s^3Y - s^2(1) - s(0) - 1$. $(s^3 - 1)Y = s^2 + 1$; factor $s^3 - 1 = (s-1)(s^2 + s + 1)$. **Answer:** $y(t) = \frac{2}{3}e^t + \frac{1}{3}e^{-t/2}\cos(\frac{\sqrt{3}}{2}t) - \frac{1}{\sqrt{3}}e^{-t/2}\sin(\frac{\sqrt{3}}{2}t)$.

15 Multiple Choice Questions

1. $\mathcal{L}\{f'(t)\}$ equals:

   (a) $F(s) - f(0)$ (b) $sF(s)$ **(c)** $sF(s) - f(0)$ (d) $s^2F(s) - f(0)$

   Explanation: Integration by parts on the definition integral yields $sF(s) - f(0)$.

2. $\mathcal{L}\{f''(t)\}$ equals:

   (a) $s^2F(s) - f'(0)$ (b) $s^2F(s) - sf(0)$ (c) $sF(s) - sf(0) - f'(0)$ **(d)** $s^2F(s) - sf(0) - f'(0)$

   Explanation: Apply the first-derivative formula twice: $\mathcal{L}\{(f')'\} = s[sF(s) - f(0)] - f'(0) = s^2F(s) - sf(0) - f'(0)$.

3. When Laplace transforms are applied to an IVP, where do the initial conditions appear?

   (a) Only at the end, when inverting $Y(s)$ **(b) In the transformed equation, automatically, as constant terms** (c) They must be imposed separately after finding $y(t)$ (d) They do not appear in the Laplace method

   Explanation: The derivative formulas contain $f(0)$, $f'(0)$, etc.; substituting their values builds the ICs directly into the algebraic equation for $Y(s)$.

4. If the ODE has order 2, the algebraic equation for $Y(s)$ in the s-domain has degree:

   (a) 1 **(b) 2** (c) 3 (d) It depends on the right-hand side

   Explanation: Applying $\mathcal{L}$ to a second-order ODE with constant coefficients always produces a polynomial equation of degree 2 in s for $Y(s)$.

5. In the s-domain, multiplying $F(s)$ by s is equivalent to what operation in the t-domain?

   (a) Integration of $f(t)$ (b) Shifting $f(t)$ **(c) Differentiating $f(t)$ (when $f(0) = 0$)** (d) Scaling $f(t)$ by a constant

   Explanation: When $f(0) = 0$, $\mathcal{L}\{f'\} = sF(s)$, so multiplication by s in the s-domain corresponds exactly to differentiation in the t-domain.

16 Common Mistakes

Common Mistakes in Laplace Transform of Derivatives

Mistake	What Students Do	Correct Approach
Writing $\mathcal{L}\{f'\} = sF(s)$	Omit the initial condition term $-f(0)$ entirely	Always write $sF(s) - f(0)$; set $f(0)$ to zero only if the IC explicitly says so
Confusing $f(0)$ and $f'(0)$ in $\mathcal{L}\{f''\}$	Write $s^2F(s) - sf'(0) - f(0)$ (positions swapped)	The correct formula is $s^2F(s) - sf(0) - f'(0)$: $f(0)$ is multiplied by s, $f'(0)$ is subtracted directly
Leaving $f(0)$, $f'(0)$ as symbols	Forget to substitute the given numerical IC values into the algebraic equation	Substitute the given numbers for $f(0)$ and $f'(0)$ immediately after writing the transformed equation
Forgetting $\mathcal{L}$ of the right-hand side	Transform only the left-hand side of the ODE, leaving the right-hand side as $g(t)$	Apply $\mathcal{L}$ to both sides; write $\mathcal{L}\{g(t)\} = G(s)$ before proceeding
Partial fraction sign errors	Copy the residue formula incorrectly, getting wrong coefficients and thus wrong $y(t)$	Always verify: multiply back by the denominator and check that numerator matches; substitute a test value of s to double-check

17 Quick Recap

Quick Recap — Topic 08: Laplace Transform of Derivatives

- $\mathcal{L}\{f'(t)\} = s\,F(s) - f(0)$ — proved by integration by parts; initial condition appears automatically.
- $\mathcal{L}\{f''(t)\} = s^2F(s) - s\,f(0) - f'(0)$ — obtained by applying the first-derivative formula twice.
- General formula: $\mathcal{L}\{f^{(n)}(t)\} = s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$.
- 4-step ODE workflow: (1) Apply $\mathcal{L}$, (2) substitute ICs, (3) solve for $Y(s)$, (4) apply $\mathcal{L}^{-1}$.
- Initial conditions are built into the transformed equation — no separate enforcement step required.
- The Laplace Transform converts calculus (differentiation) into algebra (multiplication by s) — this is the core "magic".
- After finding $Y(s)$, invert using partial fractions together with the standard Laplace table (topics covered in Unit V).
- Validity requires f continuous, $f^{(n)}$ piecewise continuous, and both of exponential order — always verify before applying.